

# POWERTRAIN CONTROL SYSTEM

## High-Level Design Document

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Classification: Synthetic Demonstration Data

## 1. System Overview

The Powertrain Control System coordinates its functions through typed AUTOSAR interfaces and ports. Software components exchange signals through provider and consumer ports.

## 2. Software Architecture

The primary software components are EngineControlManager, ThrottleControl, FuelInjectionControl, TransmissionControl, and CrankSensorManager.

## 3. Software Components

| Component            | Role        | Primary Responsibility                        |
|----------------------|-------------|-----------------------------------------------|
| EngineControlManager | Coordinator | Coordinates torque requests and engine state. |
| ThrottleControl      | Provider    | Measures throttle position and reports it.    |
| FuelInjectionControl | Consumer    | Meters fuel based on load and speed.          |
| TransmissionControl  | Consumer    | Selects gears using engine speed and torque.  |
| CrankSensorManager   | Provider    | Derives engine speed from the crank sensor.   |

## 4. Interfaces

| Interface         | Provider             | Consumer             | Signals          |
|-------------------|----------------------|----------------------|------------------|
| IThrottlePosition | ThrottleControl      | EngineControlManager | ThrottlePosition |
| IEngineSpeed      | CrankSensorManager   | EngineControlManager | EngineSpeed      |
| IFuelDemand       | EngineControlManager | FuelInjectionControl | FuelDemand       |
| IGearRequest      | EngineControlManager | TransmissionControl  | GearRequest      |
| ITorqueLimit      | TransmissionControl  | EngineControlManager | TorqueLimit      |

## 5. Ports

| Component            | Port                 | Direction | Interface         |
|----------------------|----------------------|-----------|-------------------|
| ThrottleControl      | ThrottlePosition_Out | P-PORT    | IThrottlePosition |
| EngineControlManager | ThrottlePosition_In  | R-PORT    | IThrottlePosition |
| CrankSensorManager   | EngineSpeed_Out      | P-PORT    | IEngineSpeed      |
| EngineControlManager | EngineSpeed_In       | R-PORT    | IEngineSpeed      |
| EngineControlManager | FuelDemand_Out       | P-PORT    | IFuelDemand       |
| FuelInjectionControl | FuelDemand_In        | R-PORT    | IFuelDemand       |
| EngineControlManager | GearRequest_Out      | P-PORT    | IGearRequest      |
| TransmissionControl  | GearRequest_In       | R-PORT    | IGearRequest      |
| TransmissionControl  | TorqueLimit_Out      | P-PORT    | ITorqueLimit      |
| EngineControlManager | TorqueLimit_In       | R-PORT    | ITorqueLimit      |

## 6. Signals

| Signal           | Data Type | Source               | Destination          |
|------------------|-----------|----------------------|----------------------|
| ThrottlePosition | uint8     | ThrottleControl      | EngineControlManager |
| EngineSpeed      | uint16    | CrankSensorManager   | EngineControlManager |
| FuelDemand       | uint16    | EngineControlManager | FuelInjectionControl |
| GearRequest      | uint8     | EngineControlManager | TransmissionControl  |
| TorqueLimit      | uint16    | TransmissionControl  | EngineControlManager |

## 7. Dependencies

| Source               | Relationship       | Target            | Reason                       |
|----------------------|--------------------|-------------------|------------------------------|
| ThrottleControl      | PROVIDES_INTERFACE | IThrottlePosition | Publishes throttle position. |
| EngineControlManager | REQUIRES_INTERFACE | IThrottlePosition | Consumes throttle position.  |
| CrankSensorManager   | PROVIDES_INTERFACE | IEngineSpeed      | Publishes engine speed.      |
| EngineControlManager | REQUIRES_INTERFACE | IEngineSpeed      | Consumes engine speed.       |
| EngineControlManager | PROVIDES_INTERFACE | IFuelDemand       | Publishes fuel demand.       |
| FuelInjectionControl | REQUIRES_INTERFACE | IFuelDemand       | Consumes fuel demand.        |
| EngineControlManager | PROVIDES_INTERFACE | IGearRequest      | Publishes gear request.      |
| TransmissionControl  | REQUIRES_INTERFACE | IGearRequest      | Consumes gear request.       |
| TransmissionControl  | PROVIDES_INTERFACE | ITorqueLimit      | Publishes torque limit.      |
| EngineControlManager | REQUIRES_INTERFACE | ITorqueLimit      | Consumes torque limit.       |

## 8. Functional Flows

### Flow F-001: Throttle Position Distribution

ThrottleControl publishes ThrottlePosition through IThrottlePosition. EngineControlManager receives the signal through ThrottlePosition\_In and updates its internal state model.

### Flow F-002: Engine Speed Distribution

CrankSensorManager publishes EngineSpeed through IEngineSpeed. EngineControlManager receives the signal through EngineSpeed\_In and updates its internal state model.

### Flow F-003: Fuel Demand Distribution

EngineControlManager publishes FuelDemand through IFuelDemand. FuelInjectionControl receives the signal through FuelDemand\_In and updates its internal state model.

### Flow F-004: Gear Request Distribution

EngineControlManager publishes GearRequest through IGearRequest. TransmissionControl receives the signal through GearRequest\_In and updates its internal state model.

### Flow F-005: Torque Limit Distribution

TransmissionControl publishes TorqueLimit through ITorqueLimit. EngineControlManager receives the signal through TorqueLimit\_In and updates its internal state model.

## 9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

### 10. Engineering Traceability

This synthetic HLD supports extraction, retrieval, validation and revision comparison experiments.